Arc length

1

a 11.5 units

b 19.6 units

2

a 10.47 cm

b 33.25 mm

c 9.42 cm

d 6.00 cm

3

a $\frac{5\pi}{6}$ radians

b $\frac{25\pi}{2}$ cm

4 $\left(20 + \frac{25\pi}{3}\right)$ cm

5

a 36.8 cm

b 16.8 cm

Sector area

6

a 463.10 m^2

b 313.01 m^2

7 39π $units^2$

8

a 50 cm^2

b 729.87 m^2

c 612.42 m^2

9

a $\frac{2\pi}{3}$

b 108π cm^2

10

a 6π cm^2

b 6π cm^2

c 6π cm^2

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Dimensions from length and area

11
a $r = 11.46$ **b** $r = 5.09$ **c** $r = 4.79$ **d** $r = 14.90$

12
a $\theta = 45.02^\circ$ **b** $\theta = 334.23^\circ$ **c** $\theta = 190.99^\circ$

13 $\theta = 1.95$ radians

14
a $\theta = 0.9$ radians **b** $\theta = 1.1$ radians

15
a $\theta = 2$ radians **b** $\frac{9}{4}\pi^2 \text{ cm}^2$

16
a $\angle JOK = 46^\circ$ **b** 90 cm^2

17
a $\theta = 1.67$ radians **b** $\theta = 96^\circ$

18
a $\theta = 104^\circ$ **b** 586 m^2

19 $\theta = 1.7$ radians

20
a $\theta = 0.8$ radians **b** $l = 16 \text{ m}$

21 $l = 11 \text{ cm}$

Applications

22 81 cm^2

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23 $317.9\ \text{m}^2$

24

a $48\pi\ \text{cm}^2$ **b** $336\pi\ \text{cm}^2$

25 $70\pi\ \text{cm}$

26

a $10\pi\ \text{cm}^2$ **b** $(5\pi + 8)\ \text{cm}$

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